Chapter 6: Orthogonal Polynomials

Mathematical Physics

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Outline

1. From Laplace to ordinary Legendre
2. Legendre polynomials
3. Associated Legendre and spherical harmonics
4. Hermite polynomials
5. Laguerre polynomials
6. Chebyshev polynomials
7. Summary

Where we are going

This chapter studies four families of **orthogonal polynomials** that arise by separating variables in Laplace, Poisson, Helmholtz, and Schrödinger problems. "Orthogonal" means a weighted inner product of distinct-degree polynomials vanishes; the interval and weight depend on the family.

- **Legendre P_n.** From Laplace in spherical coordinates to the ordinary ($m = 0$) Legendre equation, series termination, generating function, recurrence, Rodrigues, orthogonality, and Fourier–Legendre expansions. Associated Legendre functions and spherical harmonics Y_n^m for $m \neq 0$, with the degree n the same integer that the axisymmetric problem picks out through $\lambda = n(n+1)$.
- **Hermite H_n.** The Hermite equation and series solution, quantum harmonic oscillator motivation, generating function, Rodrigues formula, and weighted orthogonality.
- **Laguerre L_n and Chebyshev T_n.** Defining ODEs, first polynomials, weights and orthogonality, recurrences, and useful formula sheets.

A comparison table at the end collects interval, weight, and ODE for each family.

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Two-dimensional Laplace equation in polar coordinates

Step 1: separate the angular and radial parts. A function satisfying Laplace's equation $\nabla^2\psi = 0$ is called a *harmonic function*. The operator ∇^2 is the Laplacian: $\psi_{xx} + \psi_{yy}$ in Cartesian coordinates. In polar coordinates r is radius and θ is angle. We use its coordinate form without deriving it:

$$\nabla^2\psi = \frac{1}{r}\,\frac{\partial}{\partial r}\left(r\,\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\,\frac{\partial^2\psi}{\partial\theta^2} = 0. \tag{6.1.1}$$

Take a separated solution $\psi(r,\theta) = u(r)v(\theta)$. Then

$$\frac{\partial\psi}{\partial r} = u'(r)v(\theta), \qquad \frac{\partial^2\psi}{\partial\theta^2} = u(r)v''(\theta),$$

so substitution into (6.1.1) gives

$$\frac{v}{r}\,\frac{d}{dr}\big(ru'\big) + \frac{u}{r^2}v'' = 0.$$

Multiplying by $r^2/(uv)$ yields

$$\frac{r(ru')'}{u} + \frac{v''}{v} = 0,$$

Two-dimensional Laplace equation in polar coordinates (cont.)

and therefore

$$-\frac{v''}{v} = \frac{r(ru')'}{u} = \lambda, \tag{6.1.2}$$

because the left member depends only on θ and the right member only on r.
The angular equation is

$$v'' + \lambda v = 0.$$

For $\lambda > 0$, write $\lambda = \mu^2$; then

$$v(\theta) = A\cos(\mu\theta) + B\sin(\mu\theta).$$

The periodicity condition $v(\theta + 2\pi) = v(\theta)$ restricts μ: since $\cos\mu\theta$ and $\sin\mu\theta$ have period $2\pi/\mu$, the condition can hold for all θ only when 2π is a whole multiple of that period, i.e. only when μ is an integer. Hence $\mu = n$ with $n \in \mathbb{Z}$. Since the modes n and $-n$ span the same real sine–cosine pair, we may take $n \geq 0$ and $\lambda = n^2$. If $\lambda = 0$, then $v = A + B\theta$, and periodicity forces $B = 0$; this is the $n = 0$ mode. If $\lambda < 0$, the angular solutions are exponentials rather than periodic functions, so no nonzero 2π-periodic mode occurs.
The radial equation is

$$r(ru')' - n^2 u = 0.$$

Two-dimensional Laplace equation in polar coordinates (cont.)

Expanding the derivative gives a Cauchy–Euler (equidimensional) equation — one in which every term carries as many powers of r as derivatives of u —

$$r^2u'' + ru' - n^2u = 0, \tag{6.1.3}$$

which is therefore solved by trial powers $u = r^p$. For $n \geq 1$, try $u = r^p$. Since

$$u' = pr^{p-1}, \qquad u'' = p(p-1)r^{p-2},$$

we obtain

$$r^2u'' + ru' - n^2u = \left[p(p-1) + p - n^2\right]r^p = (p^2 - n^2)r^p.$$

Thus $p = \pm n$, and

$$u_n(r) = C_nr^n + D_nr^{-n} \qquad (n \geq 1).$$

For $n = 0$, (6.1.3) becomes $(ru')' = 0$. Integrating twice,

$$ru' = C_0, \qquad u' = \frac{C_0}{r}, \qquad u(r) = D_0 + C_0 \log r.$$

Two-dimensional Laplace equation in polar coordinates (cont.)

On an annulus, all these radial solutions are allowed. Inside a disk we must also examine the center. Inside a disk containing $r = 0$, boundedness removes $\log r$ and all r^{-n} terms:

$$\psi(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^n \left[A_n \cos(n\theta) + C_n \sin(n\theta) \right]. \tag{6.1.4}$$

Equivalently, with $z = re^{i\theta}$, $c_0 = A_0$, and $c_n = A_n - iC_n$ for $n \geq 1$,

$$\psi(r, \theta) = \mathrm{Re}\left(\sum_{n=0}^{\infty} c_n z^n \right), \qquad z^n = r^n e^{in\theta}.$$

Thus the regular real harmonic functions are real parts of analytic power series; exterior or annular domains additionally allow negative powers and, for the angularly independent $n = 0$ mode, a logarithm.

Three-dimensional Laplace equation in spherical coordinates

In spherical coordinates r is radius, θ is the polar angle measured from the positive vertical axis, and ϕ is the azimuthal angle around that axis. We use the spherical Laplacian without deriving this coordinate formula:

$$\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2} = 0. \tag{6.1.5}$$

Step 1: separate the radial variable. Set $\psi(r,\theta,\phi) = u(r)y(\theta,\phi)$. Because y is independent of r and u is independent of the angles,

$$\frac{\partial}{\partial r}(r^2\psi_r) = y(r^2u')',$$
$$\frac{\partial}{\partial\theta}(\sin\theta\,\psi_\theta) = u(\sin\theta\,y_\theta)_\theta.$$

Substitution into (6.1.5) and multiplication by $r^2/(uy)$ give

$$\frac{(r^2u')'}{u} = -\frac{\dfrac{1}{\sin\theta}(\sin\theta\,y_\theta)_\theta + \dfrac{1}{\sin^2\theta}y_{\phi\phi}}{y} = \lambda. \tag{6.1.6}$$

The first expression depends only on r and the second only on (θ, ϕ), so each equals a constant λ.

Step 2: radial Euler equation. Expanding $(r^2 u')' = r^2 u'' + 2ru'$ gives

$$r^2 u'' + 2ru' - \lambda u = 0.$$

Try $u = r^\sigma$. Then $u' = \sigma r^{\sigma-1}$, $u'' = \sigma(\sigma-1)r^{\sigma-2}$, and

$$\Big[\sigma(\sigma+1) - \lambda\Big] r^\sigma = 0.$$

So $\lambda = \sigma(\sigma+1)$. Once the angular problem forces $\lambda = n(n+1)$ with $n = 0, 1, 2, \ldots$ (next frames), the two exponents are

$$\sigma = n, \qquad \sigma = -n-1,$$

and

$$u_n(r) = A_n r^n + B_n r^{-n-1}. \tag{6.1.7}$$

The r^n branch is regular at the origin; the r^{-n-1} branch decays at infinity.

Three-dimensional Laplace equation in spherical coordinates (cont.)

The remaining angular eigenproblem is

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\,\frac{\partial y}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2 y}{\partial\phi^2}+\lambda y=0.$$

We first treat the **axisymmetric** case (no ϕ-dependence), which yields the ordinary Legendre equation and $\lambda = n(n+1)$. Associated Legendre modes for $m \neq 0$ come later as a short extension.

Axisymmetry: the ordinary Legendre equation

Step 1: assume no azimuthal dependence and change variables. If ψ is independent of ϕ (axisymmetry about the polar axis), the angular equation reduces to

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\,\frac{dv}{d\theta}\right) + \lambda v = 0.$$

Set $x = \cos\theta$, so $dx/d\theta = -\sin\theta$ and $d/d\theta = -\sin\theta\, d/dx$. Then

$$\sin\theta\,\frac{dv}{d\theta} = -(1-x^2)v_x,$$
$$\frac{d}{d\theta}\left(\sin\theta\,\frac{dv}{d\theta}\right) = \sin\theta\frac{d}{dx}\left[(1-x^2)v_x\right],$$

and therefore

$$\frac{d}{dx}\left[(1-x^2)v'(x)\right] + \lambda v(x) = 0. \tag{6.1.8}$$

Expanding the derivative gives the equivalent form

$$(1-x^2)v'' - 2xv' + \lambda v = 0. \tag{6.1.9}$$

(The poles $\theta = 0, \pi$ correspond to the endpoints $x = \pm 1$.)

Step 2: power series and coefficient recurrence. Set

$$v(x) = \sum_{j=0}^{\infty} a_j x^j.$$

Then

$$\begin{aligned} v' &= \sum_{j=0}^{\infty} (j+1) a_{j+1} x^j, \\ v'' &= \sum_{j=0}^{\infty} (j+2)(j+1) a_{j+2} x^j, \\ -x^2 v'' &= -\sum_{j=0}^{\infty} j(j-1) a_j x^j, \\ -2xv' &= -2 \sum_{j=0}^{\infty} j\, a_j x^j. \end{aligned}$$

Axisymmetry: the ordinary Legendre equation (cont.)

Substitution into (6.1.9) gives

$$0 = \sum_{j=0}^{\infty} [(j+2)(j+1)a_{j+2} - j(j-1)a_j - 2ja_j + \lambda a_j]\, x^j$$
$$= \sum_{j=0}^{\infty} \Big[(j+2)(j+1)a_{j+2} + \big(\lambda - j(j+1)\big)\, a_j\Big]\, x^j.$$

Hence the coefficient recurrence is

$$a_{j+2} = -\frac{\lambda - j(j+1)}{(j+2)(j+1)}\, a_j. \tag{6.1.10}$$

Even and odd chains are independent.
Step 3: select solutions that remain finite at the poles. The points $x = \pm 1$ correspond to the north and south poles of the sphere. A polynomial is finite at both. The recurrence stops at degree n precisely when $\lambda = n(n+1)$; this is the simplest way to construct the acceptable solutions.
We use the standard endpoint result without its proof: a nonzero solution finite at both endpoints exists only for these values of λ, and is a multiple of that polynomial. The following example explains why convergence only inside $(-1, 1)$ is not enough.

Axisymmetry: the ordinary Legendre equation (cont.)

Example. Take $\lambda = 2$ and start the *even* chain at $a_0 = 1$. Then (6.1.10) gives

$$a_2 = -\frac{2-0}{2\cdot 1} = -1, \qquad a_4 = -\frac{2-6}{4\cdot 3}(-1) = -\frac{1}{3}, \qquad a_6 = -\frac{2-20}{6\cdot 5}\left(-\frac{1}{3}\right) = -\frac{1}{5},$$

and the pattern is $a_{2k} = -1/(2k-1)$ for $k \geq 1$. Subtracting the Taylor series for $\log(1+x)$ and $\log(1-x)$ gives $\sum_{k\geq 1} x^{2k-1}/(2k-1) = \frac{1}{2}\log[(1+x)/(1-x)]$ for $|x| < 1$. Hence

$$v(x) = 1 - \sum_{k\geq 1} \frac{x^{2k}}{2k-1} = 1 - x\sum_{k\geq 1} \frac{x^{2k-1}}{2k-1} = 1 - \frac{x}{2}\log\frac{1+x}{1-x}.$$

This is a genuine solution of (6.1.9) with $\lambda = 2$ (a one-line check by substitution), finite at every interior point, yet $v(x) \to -\infty$ as $x \to 1^-$: useless on the sphere. Meanwhile the *odd* chain with $\lambda = 2$ terminates immediately and gives the polynomial $v = x$.
For the allowed value $\lambda = n(n+1)$, the recurrence factors as

$$a_{j+2} = -\frac{(n-j)(n+j+1)}{(j+2)(j+1)}\, a_j. \tag{6.1.11}$$

Axisymmetry: the ordinary Legendre equation (cont.)

At $j = n$ one has $a_{n+2} = 0$, so the same-parity chain is a polynomial of degree n. Set the initial coefficient of the opposite-parity chain to zero. The surviving polynomial is bounded at both poles.
With $\lambda = n(n+1)$, equations (6.1.8)–(6.1.9) become the ordinary Legendre equation

$$\frac{d}{dx}\left[(1-x^2)v'(x)\right] + n(n+1)v(x) = 0, \tag{6.1.12}$$

or

$$(1-x^2)v'' - 2xv' + n(n+1)v = 0. \tag{6.1.13}$$

Choose the scale so that $P_n(1) = 1$; the resulting polynomial is the *Legendre polynomial* P_n. Since the recurrence advances by two and only the terminating parity chain is retained, its powers are $x^n, x^{n-2}, \ldots$. Thus

$$P_n(-x) = (-1)^n P_n(x).$$

Rodrigues' formula below supplies the normalization directly. The generating function below fixes all the normalized coefficients explicitly.

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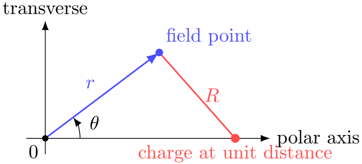
Generating function from the Coulomb potential

Step 1: write the axisymmetric potential. Place a point charge one unit from the origin on the positive polar axis. For a field point with spherical coordinates (r, θ, ϕ), the law of cosines gives

$$R^2 = r^2 + 1^2 - 2r(1)\cos\theta = 1 - 2r\cos\theta + r^2,$$

so, for a nonzero physical scale factor C, the Coulomb potential is

$$\varphi(r,\theta) = \frac{C}{R} = \frac{C}{\sqrt{1 - 2r\cos\theta + r^2}}. \tag{6.2.1}$$



$R^2 = 1 + r^2 - 2r\cos\theta$

Generating function from the Coulomb potential (cont.)

In the picture the horizontal line is the polar axis $\theta = 0$, along which the charge sits, and θ is measured from it exactly as in (6.1.5). That axis is a genuine Cartesian direction; do not confuse it with the variable $x = \cos\theta$ used in the formulas of this chapter, which is the cosine of that angle and ranges over $[-1, 1]$.

For $r < 1$ the source is outside the region under study. The potential is harmonic, finite at the origin, and independent of ϕ. Our separated solutions therefore suggest the expansion

$$\frac{1}{\sqrt{1 - 2r\cos\theta + r^2}} = \sum_{n=0}^{\infty} a_n r^n P_n(\cos\theta). \tag{6.2.2}$$

This is a physical motivation for the generating function. We quote the existence of this convergent expansion for $r < 1$; the axis calculation now fixes its coefficients.

Step 2: determine the coefficients on the axis. Set $\theta = 0$. Then $\cos\theta = 1$, $P_n(1) = 1$, and

$$1 - 2r\cos 0 + r^2 = (1 - r)^2.$$

Because $0 \le r < 1$, $|1 - r| = 1 - r$, so (6.2.2) becomes

$$\frac{1}{1 - r} = \sum_{n=0}^{\infty} a_n r^n = \sum_{n=0}^{\infty} r^n \qquad (|r| < 1).$$

Generating function from the Coulomb potential (cont.)

Uniqueness of power-series coefficients gives $a_n = 1$. Writing $t = r$ and $x = \cos\theta$,

$$g(x,t) = \frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n, \qquad |t|<1, \quad -1 \le x \le 1. \tag{6.2.3}$$

Step 3: expand the first three P_n from the generating function. For small t, write

$$g(x,t) = (1-u)^{-1/2}, \qquad u = 2xt - t^2,$$

and use the binomial series

$$(1-u)^{-1/2} = 1 + \tfrac{1}{2}u + \tfrac{1}{2}\cdot\tfrac{3}{2}\frac{u^2}{2!} + O(u^3) = 1 + \tfrac{1}{2}u + \tfrac{3}{8}u^2 + O(u^3).$$

Collect powers of t up to t^2:

$$\begin{aligned} u &= 2xt - t^2, \\ u^2 &= (2xt - t^2)^2 = 4x^2t^2 + O(t^3). \end{aligned}$$

Generating function from the Coulomb potential (cont.)

Hence

$$\begin{aligned} g &= 1 + \tfrac{1}{2}(2xt - t^2) + \tfrac{3}{8}(4x^2t^2) + O(t^3) \\ &= 1 + xt + \left(-\tfrac{1}{2} + \tfrac{3}{2}x^2\right)t^2 + O(t^3) \\ &= 1 + xt + \tfrac{1}{2}(3x^2 - 1)\,t^2 + O(t^3). \end{aligned}$$

Matching $\sum P_n(x)t^n$ therefore gives the first three polynomials explicitly:

$$P_0(x) = 1, \qquad P_1(x) = x, \qquad P_2(x) = \frac{1}{2}(3x^2 - 1). \tag{6.2.4}$$

Checks: $P_n(1) = 1$ and $P_n(-x) = (-1)^n P_n(x)$ hold for $n = 0, 1, 2$.

Three-term recurrence

Step 1: differentiate the generating function. Let

$$D(x,t) = 1 - 2xt + t^2, \qquad g(x,t) = D(x,t)^{-1/2}.$$

Differentiating with respect to t gives

$$\frac{\partial g}{\partial t} = -\frac{1}{2} D^{-3/2} \frac{\partial D}{\partial t} = \frac{x-t}{D^{3/2}}.$$

Termwise differentiation of the power series gives

$$\frac{\partial g}{\partial t} = \sum_{n=1}^{\infty} n P_n(x) t^{n-1}. \tag{6.2.5}$$

Multiplying $g_t = (x-t)D^{-3/2}$ by D gives $(x-t)g = D g_t$. Substituting both series,

$$(x-t)\sum_{n=0}^{\infty} P_n t^n = (1 - 2xt + t^2) \sum_{n=1}^{\infty} n P_n t^{n-1},$$

where $P_n = P_n(x)$. Expanding,

$$\sum_{n=0}^{\infty} x P_n t^n - \sum_{n=0}^{\infty} P_n t^{n+1} = \sum_{n=1}^{\infty} n P_n t^{n-1} - 2\sum_{n=1}^{\infty} n x P_n t^n + \sum_{n=1}^{\infty} n P_n t^{n+1}. \tag{6.2.6}$$

Three-term recurrence (cont.)

Step 2: compare coefficients of t^n. With the convention $P_{-1} = 0$, the left coefficient of t^n is $xP_n - P_{n-1}$. On the right,

$$[t^n] \sum_{j \geq 1} jP_j t^{j-1} = (n+1)P_{n+1}, \qquad [t^n]\Big(-2 \sum_{j \geq 1} jxP_j t^j\Big) = -2nxP_n,$$

and

$$[t^n] \sum_{j \geq 1} jP_j t^{j+1} = (n-1)P_{n-1}$$

(using $P_{-1} = 0$). Equating coefficients,

$$xP_n - P_{n-1} = (n+1)P_{n+1} - 2nxP_n + (n-1)P_{n-1}.$$

Collecting the P_n and P_{n-1} terms gives

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x), \qquad n \geq 0, \tag{6.2.7}$$

with $P_{-1} = 0$ and $P_0 = 1$. For $n = 0$, $P_1 = xP_0 = x$.

First few Legendre polynomials

We already obtained $P_0 = 1$, $P_1 = x$, and $P_2 = (3x^2 - 1)/2$ from the generating function. The recurrence (6.2.7) supplies the next one without another series expansion:

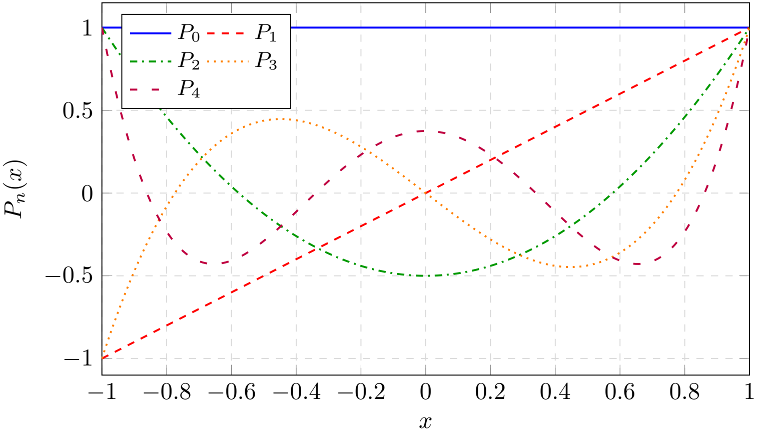
$$3P_3 = 5xP_2 - 2P_1 = \frac{5x(3x^2 - 1)}{2} - 2x = \frac{15x^3 - 9x}{2}.$$

Thus

$$P_3(x) = \frac{1}{2}(5x^3 - 3x).$$

It has odd parity and $P_3(1) = 1$, as expected. The following plot shows how the oscillations change with degree.

Shapes of the first Legendre polynomials



Each P_n is bounded on $[-1, 1]$ and normalized by $P_n(1) = 1$. Increasing n adds interior zeros while preserving $\int_{-1}^{1} P_m(x)P_n(x)\,dx = 0$ for $m \neq n$.

Rodrigues' formula

The recurrence generates $P_0, P_1, P_2, \ldots$ in order. A complementary formula constructs any one polynomial directly:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n. \tag{6.2.8}$$

This is Rodrigues' formula. We use the identity without its general proof. For $n = 2$, the calculation is just polynomial differentiation:

$$\begin{aligned} (x^2 - 1)^2 &= x^4 - 2x^2 + 1, \\ \frac{d}{dx}(x^2 - 1)^2 &= 4x^3 - 4x, \\ \frac{d^2}{dx^2}(x^2 - 1)^2 &= 12x^2 - 4, \\ P_2(x) &= \frac{12x^2 - 4}{2^2\, 2!} = \frac{3x^2 - 1}{2}. \end{aligned}$$

The factor $2^n n!$ makes $P_n(1) = 1$: in $(x - 1)^n (x + 1)^n$, the only term surviving after n derivatives at $x = 1$ differentiates $(x - 1)^n$ all n times and equals $n!\, 2^n$.

Rodrigues' formula (cont.)

Two consequences needed for the norm. The highest power in $(x^2-1)^n$ is x^{2n}. Hence

$$\frac{d^n}{dx^n}x^{2n} = \frac{(2n)!}{n!}x^n,$$

and division by $2^n n!$ gives

$$P_n(x) = \frac{(2n)!}{2^n(n!)^2}x^n + \text{lower powers}. \tag{6.2.9}$$

Differentiating n more times removes all lower powers:

$$P_n^{(n)}(x) = \frac{(2n)!}{2^n n!}. \tag{6.2.10}$$

The zeros of order n in $(x^2-1)^n$ at $x=\pm 1$ also make the boundary terms vanish when we integrate by parts up to n times. These are the two facts used next.

Connection to Chapter 1. Cauchy's derivative formula applied to $(t^2-1)^n$ also writes Rodrigues' formula as

$$P_n(x) = \frac{1}{2^{n+1}\pi i}\oint_C \frac{(t^2-1)^n}{(t-x)^{n+1}}\,dt, \tag{6.2.11}$$

where C is a simple counterclockwise contour enclosing x. We will use the derivative form for the norm calculation.

Orthogonality and norm of Legendre polynomials

The Legendre equation is

$$\frac{d}{dx}\left[(1-x^2)P_n'(x)\right] + n(n+1)P_n(x) = 0. \tag{6.2.12}$$

This is the Chapter 4 problem with $p = 1 - x^2$, $q = 0$, and $w = 1$. At either endpoint,

$$(1-x^2)(P_n P_m' - P_n' P_m) \longrightarrow 0,$$

since polynomials and their derivatives remain finite. The eigenvalues $n(n+1)$ differ for distinct nonnegative degrees. Therefore Chapter 4's weighted orthogonality result gives

$$\int_{-1}^{1} P_n(x)P_m(x)\,dx = 0 \qquad (n \neq m). \tag{6.2.13}$$

Orthogonality and norm of Legendre polynomials (cont.)

Norm via Rodrigues and integration by parts. For $n = 0$, $P_0 = 1$ gives $I_0 = 2$ directly. For $n \geq 1$, set

$$I_n := \int_{-1}^{1} P_n(x)^2 \, dx = \frac{1}{2^n n!} \int_{-1}^{1} P_n(x) \frac{d^n}{dx^n}(x^2 - 1)^n \, dx.$$

Integrate by parts once:

$$\begin{aligned} I_n = \frac{1}{2^n n!} \left[P_n \frac{d^{n-1}}{dx^{n-1}}(x^2 - 1)^n \right]_{-1}^{1} \\ - \frac{1}{2^n n!} \int_{-1}^{1} P_n' \frac{d^{n-1}}{dx^{n-1}}(x^2 - 1)^n \, dx. \end{aligned}$$

Because $(x^2 - 1)^n = (x-1)^n (x+1)^n$ has a zero of order n at each endpoint,

$$\left. \frac{d^j}{dx^j}(x^2 - 1)^n \right|_{x = \pm 1} = 0, \qquad 0 \leq j \leq n - 1.$$

Orthogonality and norm of Legendre polynomials (cont.)

Thus every boundary term in n successive integrations by parts vanishes, and

$$I_n = \frac{(-1)^n}{2^n n!} \int_{-1}^{1} P_n^{(n)}(x)(x^2-1)^n \, dx. \tag{6.2.14}$$

Use $P_n^{(n)} = (2n)!/(2^n n!)$ from (6.2.10) and $(-1)^n(x^2-1)^n = (1-x^2)^n$:

$$I_n = \frac{(2n)!}{2^{2n}(n!)^2} \int_{-1}^{1} (1-x^2)^n \, dx.$$

Orthogonality and norm of Legendre polynomials (cont.)

To evaluate the remaining integral, set $x = 2u - 1$. Then $dx = 2\,du$ and $1 - x^2 = 4u(1-u)$, so with the beta function $B(a,b) = \int_0^1 u^{a-1}(1-u)^{b-1}\,du = \Gamma(a)\Gamma(b)/\Gamma(a+b)$,

$$\int_{-1}^{1} (1-x^2)^n\,dx = 2\cdot 4^n \int_0^1 u^n(1-u)^n\,du = 2^{2n+1} B(n+1, n+1)$$
$$= 2^{2n+1}\frac{(n!)^2}{(2n+1)!}.$$

Therefore

$$I_n = \frac{(2n)!}{2^{2n}(n!)^2}\cdot\frac{2^{2n+1}(n!)^2}{(2n+1)!} = \frac{2}{2n+1}.$$

Combining orthogonality and norm,

$$\int_{-1}^{1} P_n(x)P_m(x)\,dx = \frac{2}{2n+1}\,\delta_{nm}. \tag{6.2.15}$$

Fourier–Legendre series

We use the standard completeness result without proof: Legendre polynomials form a complete orthogonal system in $L^2([-1,1])$, just as the sines did in Chapter 4. With projection coefficients, their partial sums converge in the L^2 norm. Thus, for $f \in L^2([-1,1])$,

$$f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$$

in the L^2 sense. Multiply by $P_m(x)$ and integrate:

$$\int_{-1}^{1} f(x) P_m(x)\, dx = \sum_{n=0}^{\infty} a_n \int_{-1}^{1} P_n(x) P_m(x)\, dx = a_m \frac{2}{2m+1}.$$

Solving for a_m gives

$$a_m = \frac{2m+1}{2} \int_{-1}^{1} f(x) P_m(x)\, dx. \tag{6.2.16}$$

Therefore the Fourier–Legendre expansion is

$$f(x) = \sum_{n=0}^{\infty} \left[\frac{2n+1}{2} \int_{-1}^{1} f(s) P_n(s)\, ds \right] P_n(x). \tag{6.2.17}$$

Useful formulas for the Legendre polynomial

differential equation	$(1-x^2)P_n'' - 2x\,P_n' + n(n+1)P_n = 0$
series solution	$P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k (2n-2k)!}{2^n k!\,(n-2k)!\,(n-k)!}\, x^{n-2k}$
generating function	$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)\, t^n$
recurrence	$(n+1)P_{n+1} = (2n+1)x\,P_n - n\,P_{n-1}$
orthogonality	$\int_{-1}^{1} P_m(x)P_n(x)\,dx = \frac{2}{2n+1}\,\delta_{nm}$
Rodrigues	$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n$
Schläfli	$P_n(x) = \frac{1}{2^{n+1}\pi i} \oint \frac{(t^2-1)^n}{(t-x)^{n+1}}\, dt \qquad (6.2.11)$

The finite sum follows by expanding Rodrigues' formula; $\lfloor n/2 \rfloor$ means the greatest integer no larger than $n/2$. Generating series: $|t| < 1$ for $x \in [-1, 1]$. The contour encloses $t = x$.

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Associated Legendre equation and Y_n^m

When the field depends on ϕ, separate $y(\theta, \phi) = v(\theta) w(\phi)$. As in the polar calculation, the separated azimuthal modes can be chosen as $w = e^{im\phi}$; periodicity $w(\phi + 2\pi) = w(\phi)$ requires $m \in \mathbb{Z}$. Return to the angular equation in §6.1 and put $y(\theta, \phi) = v(\theta) e^{im\phi}$:

$$\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \, \frac{\partial y}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2 y}{\partial\phi^2} + \lambda y = 0.$$

Since $\dfrac{\partial^2}{\partial\phi^2} e^{im\phi} = (im)^2 e^{im\phi} = -m^2 e^{im\phi}$, every term carries the common factor $e^{im\phi}$, which is never zero and therefore cancels:

$$\frac{1}{\sin\theta} \frac{d}{d\theta} \left(\sin\theta \, \frac{dv}{d\theta} \right) + \left[\lambda - \frac{m^2}{\sin^2\theta} \right] v = 0.$$

Now set $x = \cos\theta$. The first term was already converted in the axisymmetric frame: there $\sin\theta \, dv/d\theta = -(1 - x^2) v'$ gave

$$\frac{1}{\sin\theta} \frac{d}{d\theta} \left(\sin\theta \, \frac{dv}{d\theta} \right) = \frac{d}{dx} \left[(1 - x^2) v'(x) \right],$$

Associated Legendre equation and Y_n^m (cont.)

and for the new term $\sin^2\theta = 1 - \cos^2\theta = 1 - x^2$, so $m^2/\sin^2\theta$ becomes $m^2/(1-x^2)$. The polar equation is therefore

$$\frac{d}{dx}\left[(1-x^2)v'(x)\right] + \left[\lambda - \frac{m^2}{1-x^2}\right] v(x) = 0. \tag{6.3.1}$$

For a smooth angular field on the whole sphere, the allowed values are $\lambda = n(n+1)$ with $n = |m|, |m|+1, \ldots$ (endpoint result quoted). The equation becomes

$$(1-x^2)v'' - 2xv' + \left[n(n+1) - \frac{m^2}{1-x^2}\right] v = 0. \tag{6.3.2}$$

Instead of repeating a long series analysis at the poles, construct these solutions from the Legendre polynomials already known.

Associated Legendre equation and Y_n^m (cont.)

A derivative construction (quoted). The associated Legendre functions on $[-1, 1]$ are obtained from the known polynomials by

$$P_n^m(x) = (-1)^m (1 - x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}, \quad m = 0, 1, \ldots, n. \tag{6.3.3}$$

The sign $(-1)^m$ is a convention, called the Condon–Shortley phase. For example, $P_1 = x$ and $P_2 = (3x^2 - 1)/2$ give

$$\begin{aligned} P_1^1(x) &= -\sqrt{1 - x^2}, \\ P_2^1(x) &= -3x\sqrt{1 - x^2}, \\ P_2^2(x) &= (1 - x^2)\frac{d^2}{dx^2}\frac{3x^2 - 1}{2} = 3(1 - x^2). \end{aligned}$$

For $m = 0$ this returns P_n. For $m > n$, the derivative is zero, so it produces no additional mode. This construction supplies the regular angular solutions selected above.

Associated Legendre equation and Y_n^m (cont.)

Spherical harmonics: name the angular modes. An unnormalized angular basis is

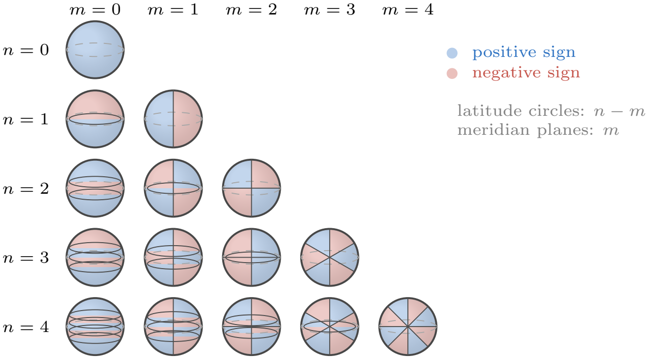
$$\mathcal{Y}_n^m(\theta,\phi) = P_n^{|m|}(\cos\theta)\, e^{im\phi}, \qquad -n \le m \le n.$$

Normalized spherical harmonics Y_n^m differ from this by a constant and a fixed phase. Many texts call the degree ℓ and write the harmonics as $Y_{\ell m}$; here the degree is called n. With the same normalization and phase convention, $Y_{\ell m}$ and Y_n^m denote the same function when $\ell = n$. Combining radial and angular factors, the regular interior expansion is

$$\psi_{\mathrm{int}}(r,\theta,\phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^{n} a_{nm} r^n \mathcal{Y}_n^m(\theta,\phi), \tag{6.3.4}$$

and the decaying exterior expansion uses r^{-n-1} instead of r^n.

Spherical-harmonic nodal patterns



Schematic counts, not exact nodal locations. Rows: degree n; columns: order m. The factor $P_n^{|m|}(\cos\theta)$ has $n - |m|$ interior nodal latitudes. For $m > 0$, a real factor $\cos(m\phi)$ or $\sin(m\phi)$ adds m meridian planes; for $m = 0$, there are none.

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The Hermite equation and series solution

Step 1: substitute a power series. The physicists' Hermite polynomials solve

$$y'' - 2xy' + 2ny = 0, \qquad n = 0, 1, 2, \ldots . \tag{6.4.1}$$

Let $y(x) = \sum_{j=0}^{\infty} a_j x^j$. Then

$$y' = \sum_{j=0}^{\infty} (j+1) a_{j+1} x^j,$$

$$y'' = \sum_{j=0}^{\infty} (j+2)(j+1) a_{j+2} x^j,$$

$$-2xy' = -2 \sum_{j=0}^{\infty} j a_j x^j.$$

Substitution into (6.4.1) gives

$$\sum_{j=0}^{\infty} \left[(j+2)(j+1) a_{j+2} + 2(n-j) a_j \right] x^j = 0.$$

The Hermite equation and series solution (cont.)

Every coefficient vanishes, so

$$a_{j+2} = \frac{2(j-n)}{(j+2)(j+1)} a_j. \tag{6.4.2}$$

Step 2: termination condition. Even and odd chains are independent. For integer $n \geq 0$, set $j = n$ in (6.4.2):

$$a_{n+2} = \frac{2(n-n)}{(n+2)(n+1)} a_n = 0.$$

Thus the same-parity chain terminates at degree n, giving a polynomial of exact degree n. The opposite-parity chain never hits a zero numerator, so we set its initial coefficient to zero (the non-polynomial second solution of (6.4.1)).
Step 3: explicit series from the downward product. Adopt the physicists' leading coefficient $a_n = 2^n$. The two-step recurrence (6.4.2) run *downward* from degree n is

$$a_j = \frac{(j+2)(j+1)}{2(j-n)} a_{j+2} \qquad (j = n-2, n-4, \ldots).$$

The Hermite equation and series solution (cont.)

The first two downward steps are

$$a_{n-2} = -\frac{n(n-1)}{4}\, a_n, \qquad a_{n-4} = -\frac{(n-2)(n-3)}{8}\, a_{n-2},$$

whenever the indicated indices are nonnegative. After k steps, the numerator contains the first $2k$ descending factors of $n!$, and the denominator is $(-4)(-8)\cdots(-4k) = (-4)^k k!$. Hence

$$a_{n-2k} = \frac{2^n n!}{(-4)^k k!(n-2k)!} = \frac{(-1)^k 2^{n-2k} n!}{k!(n-2k)!}.$$

Writing $a_{n-2k}x^{n-2k} = (-1)^k (2x)^{n-2k}\, n!/\big(k!\,(n-2k)!\big)$ and summing gives

$$H_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k (2x)^{n-2k} \frac{n!}{k!(n-2k)!}. \tag{6.4.3}$$

Low-n checks:

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x.$$

The Hermite equation and series solution (cont.)

Multiplying (6.4.1) by e^{-x^2} yields the Sturm–Liouville form

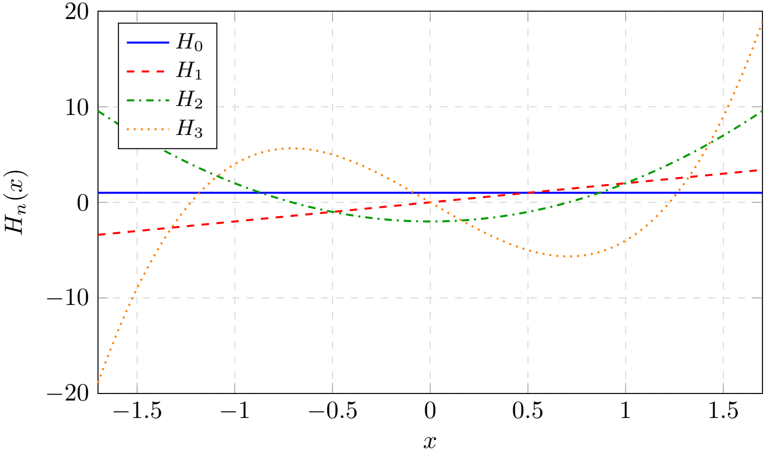
$$\frac{d}{dx}\left(e^{-x^2}y'\right) + 2ne^{-x^2}y = 0. \tag{6.4.4}$$

The same Lagrange-identity argument used for Legendre shows, for $n \neq m$,

$$\int_{-\infty}^{\infty} H_n(x)H_m(x)e^{-x^2}\,dx = 0. \tag{6.4.5}$$

(The diagonal norm is computed from the generating function below.)

Shapes of the first Hermite polynomials



Hermite polynomials grow away from the origin; the physical oscillator functions are $e^{-x^2/2}H_n(x)$.

Quantum harmonic oscillator

For a mass-M particle in $V(x) = \frac{1}{2}M\omega^2x^2$, the time-independent Schrödinger equation is given below. Here E is the energy, $\omega > 0$ is the oscillator's angular frequency, $\hbar$ is the reduced Planck constant, and ψ is the spatial wavefunction:

$$-\frac{\hbar^2}{2M}\psi'' + \frac{1}{2}M\omega^2x^2\psi = E\psi.$$

Use $\xi = \sqrt{M\omega/\hbar}\,x$. Then

$$\frac{d^2\psi}{dx^2} = \frac{M\omega}{\hbar}\frac{d^2\psi}{d\xi^2}, \qquad \frac{1}{2}M\omega^2x^2 = \frac{1}{2}\hbar\omega\xi^2.$$

Substitution and division by $\hbar\omega/2$ give $-\psi_{\xi\xi} + \xi^2\psi = (2E/\hbar\omega)\psi$, or

$$\frac{d^2\psi}{d\xi^2} + \left(\varepsilon - \xi^2\right)\psi = 0, \qquad \varepsilon = \frac{2E}{\hbar\omega}. \tag{6.4.6}$$

For large $|\xi|$, $\psi'' - \xi^2\psi \approx 0$ suggests the decaying factor $e^{-\xi^2/2}$. Set

$$\psi(\xi) = e^{-\xi^2/2}\,y(\xi).$$

Quantum harmonic oscillator (cont.)

Compute the derivatives by the product rule. Write $f := e^{-\xi^2/2}$, so $f' = -\xi f$ and $f'' = (-1+\xi^2)f$ (because $f'' = -(f + \xi f') = -(f + \xi(-\xi f)) = (-1+\xi^2)f$). Then

$$\begin{aligned}
\psi' &= f'y + fy' = e^{-\xi^2/2}\big(y' - \xi y\big),\\
\psi'' &= f''y + 2f'y' + fy''\\
&= e^{-\xi^2/2}\Big[(-1+\xi^2)y + 2(-\xi)y' + y''\Big]\\
&= e^{-\xi^2/2}\big(y'' - 2\xi y' + (\xi^2 - 1)y\big).
\end{aligned}$$

Substitute into (6.4.6):

$$\psi'' + (\varepsilon - \xi^2)\psi = e^{-\xi^2/2}\Big[y'' - 2\xi y' + (\xi^2 - 1)y + (\varepsilon - \xi^2)y\Big] = e^{-\xi^2/2}\big[y'' - 2\xi y' + (\varepsilon - 1)y\big].$$

The Gaussian prefactor never vanishes, so

$$y'' - 2\xi y' + (\varepsilon - 1)y = 0. \tag{6.4.7}$$

A power series $y = \sum a_j \xi^j$ has coefficient recurrence

$$a_{j+2} = \frac{2j + 1 - \varepsilon}{(j+2)(j+1)}\, a_j.$$

Quantum harmonic oscillator (cont.)

Why choose a terminating series? A bound state requires $\int_{-\infty}^{\infty} |\psi|^2 \, d\xi < \infty$. A polynomial y times $e^{-\xi^2/2}$ satisfies this requirement. In contrast, a nonterminating parity chain has

$$\frac{a_{j+2}}{a_j} \sim \frac{2}{j},$$

like the coefficients of e^{ξ^2}. This suggests a growing exponential component that the prefactor $e^{-\xi^2/2}$ cannot suppress. This coefficient comparison is intuition, not a full asymptotic proof. We use the standard bound-state result: decay at both ends selects a terminating chain, with the other parity chain absent.
Termination at degree n means the numerator $2j + 1 - \varepsilon$ vanishes at $j = n$, i.e.

$$\varepsilon = 2n + 1, \qquad n = 0, 1, 2, \dots,$$

and then $\varepsilon - 1 = 2n$, so (6.4.7) is exactly the Hermite equation $y'' - 2\xi y' + 2ny = 0$. Since $\varepsilon = 2E/(\hbar\omega)$, the energies are therefore

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \qquad n = 0, 1, 2, \dots, \tag{6.4.8}$$

and the spatial factors are $\psi_n \propto e^{-\xi^2/2} H_n(\xi)$.

Generating function, norm, and Rodrigues

Step 1: generating function from the series. Define

$$g(x,t) := \exp(2xt - t^2) = e^{2xt} e^{-t^2}.$$

Expand each exponential:

$$e^{2xt} = \sum_{m=0}^{\infty} \frac{(2x)^m}{m!}\, t^m, \qquad e^{-t^2} = \sum_{j=0}^{\infty} \frac{(-1)^j}{j!}\, t^{2j}.$$

The product is a double sum. The coefficient of t^n arises from pairs (m,j) with $m + 2j = n$, i.e. $m = n - 2j$ for $j = 0, \dots, \lfloor n/2 \rfloor$:

$$\begin{aligned} [t^n] g(x,t) &= \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(2x)^{n-2j}}{(n-2j)!} \cdot \frac{(-1)^j}{j!} \\ &= \frac{1}{n!} \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j (2x)^{n-2j} \frac{n!}{j!\,(n-2j)!}. \end{aligned}$$

Generating function, norm, and Rodrigues (cont.)

Comparing with the series solution (6.4.3), the sum in brackets is exactly $H_n(x)$. Therefore

$$g(x,t) = e^{-t^2+2xt} = \sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!}. \tag{6.4.9}$$

Step 2: Rodrigues from $g = e^{x^2} e^{-(x-t)^2}$. Rewrite the exponent:

$$2xt - t^2 = x^2 - (x-t)^2, \qquad g(x,t) = e^{x^2}\, e^{-(x-t)^2}.$$

Taylor-expand in t about $t = 0$:

$$e^{-(x-t)^2} = \sum_{n=0}^{\infty} \frac{t^n}{n!}\, \frac{\partial^n}{\partial t^n} e^{-(x-t)^2}\bigg|_{t=0}.$$

On the composite $e^{-(x-t)^2}$, one has $\partial/\partial t = -\partial/\partial x$, so

$$\frac{\partial^n}{\partial t^n} e^{-(x-t)^2}\bigg|_{t=0} = (-1)^n \frac{d^n}{dx^n} e^{-x^2}.$$

Generating function, norm, and Rodrigues (cont.)

Hence

$$g(x,t) = e^{x^2} \sum_{n=0}^{\infty} \frac{t^n}{n!} (-1)^n \frac{d^n}{dx^n} e^{-x^2} = \sum_{n=0}^{\infty} \left[(-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \right] \frac{t^n}{n!}.$$

Matching coefficients with (6.4.9) gives Rodrigues' formula

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}. \tag{6.4.10}$$

Step 3: recurrences from $\partial_t g$ and $\partial_x g$. Differentiate the closed form:

$$\frac{\partial g}{\partial t} = (2x - 2t)g, \qquad \frac{\partial g}{\partial x} = 2tg.$$

Using the series (6.4.9),

$$\partial_t g = \sum_{n=1}^{\infty} H_n \frac{t^{n-1}}{(n-1)!} = \sum_{n=0}^{\infty} H_{n+1} \frac{t^n}{n!}.$$

Multiplication by t shifts the index in the other term:

$$2tg = \sum_{n=1}^{\infty} 2H_{n-1} \frac{t^n}{(n-1)!} = \sum_{n=1}^{\infty} 2nH_{n-1} \frac{t^n}{n!},$$

Generating function, norm, and Rodrigues (cont.)

where $1/(n-1)! = n/n!$. Thus the coefficient of $t^n/n!$ in $(2x-2t)g$ is $2xH_n - 2nH_{n-1}$. Comparing with $\partial_t g$ gives, for $n \geq 1$,

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x). \tag{6.4.11}$$

Likewise, $\partial_x g = 2tg$. The same power shift just used gives

$$\sum_{n=0}^{\infty} H_n'(x)\frac{t^n}{n!} = \sum_{n=1}^{\infty} 2nH_{n-1}(x)\frac{t^n}{n!}.$$

Comparing coefficients of $t^n/n!$ gives

$$H_n'(x) = 2nH_{n-1}(x), \qquad n \geq 1. \tag{6.4.12}$$

The constant coefficient separately gives $H_0' = 0$.

Step 4: norm via the double generating function. For real s, t near 0,

$$I(s,t) := \int_{-\infty}^{\infty} e^{-x^2} g(x,s)g(x,t)\,dx = \int_{-\infty}^{\infty} e^{-x^2-s^2+2sx-t^2+2tx}\,dx.$$

Complete the square in x:

$$-x^2 + 2(s+t)x - s^2 - t^2 = -\big[x-(s+t)\big]^2 + 2st,$$

so

$$I(s,t) = \sqrt{\pi}\,e^{2st} = \sqrt{\pi}\sum_{k=0}^{\infty}\frac{(2st)^k}{k!} = \sqrt{\pi}\sum_{k=0}^{\infty}\frac{2^k}{k!}\,s^k t^k.$$

On the other hand, expand both generating functions:

$$g(x,s)g(x,t) = \sum_{m,n=0}^{\infty} H_m(x)H_n(x)\frac{s^m t^n}{m!\,n!},$$

For s, t near zero, the Gaussian decay permits termwise integration:

$$I(s,t) = \sum_{m,n=0}^{\infty}\left(\int_{-\infty}^{\infty} H_m(x)H_n(x)e^{-x^2}\,dx\right)\frac{s^m t^n}{m!\,n!}.$$

Generating function, norm, and Rodrigues (cont.)

The right-hand closed form $\sqrt{\pi}\, e^{2st}$ contains only diagonal powers $s^k t^k$. Equating coefficients of $s^m t^n$:

- if $m \neq n$, the right side has coefficient 0, so $\displaystyle\int H_m H_n e^{-x^2}\, dx = 0$;
- if $m = n$, compare $[s^n t^n]$:

$$\frac{1}{(n!)^2} \int_{-\infty}^{\infty} H_n^2 e^{-x^2}\, dx = \sqrt{\pi}\, \frac{2^n}{n!},$$

  Multiply by $(n!)^2$ to obtain the diagonal norm.

Combining both cases,

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2}\, dx = 2^n n! \sqrt{\pi}\, \delta_{mn}. \tag{6.4.13}$$

Low-n check: $H_0 = 1$ and $\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}$.

Useful formulas for the Hermite polynomial

differential equation	$H_n'' - 2x\, H_n' + 2n\, H_n = 0$
series solution	$H_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k (2x)^{n-2k} \dfrac{n!}{k!\,(n-2k)!}$
generating function	$e^{-t^2+2tx} = \sum_{n=0}^{\infty} H_n(x)\, \dfrac{t^n}{n!}$
recurrence	$H_n' = 2n\, H_{n-1}, \quad H_{n+1} = 2x\, H_n - 2n\, H_{n-1}$
orthogonality	$\int_{-\infty}^{\infty} H_m H_n\, e^{-x^2} dx = n!\, 2^n \sqrt{\pi}\, \delta_{mn}$
Rodrigues	$H_n(x) = (-1)^n e^{x^2} \dfrac{d^n}{dx^n} e^{-x^2}$

Here $n \geq 0$; the displayed recurrences use $n \geq 1$, with $H_0 = 1$, $H_1 = 2x$.

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Laguerre equation, polynomials, and orthogonality

Standard Laguerre polynomials solve

$$x\,y'' + (1-x)y' + n\,y = 0, \qquad n = 0, 1, 2, \dots. \tag{6.5.1}$$

Generalized Laguerre polynomials occur in the radial hydrogen-atom equation. Here we study only the standard family, defined by (6.5.1) and normalized by $L_n(0) = 1$.

Step 1: power series and coefficient recurrence. Dividing by x gives $P = (1-x)/x$ and $Q = n/x$, so $\lim_{x\to 0} xP = 1$ and $\lim_{x\to 0} x^2 Q = 0$. The Chapter 3 indicial test gives $r(r-1) + r = r^2 = 0$. The finite branch is therefore an ordinary power series; the second solution has a logarithmic singularity. We keep the branch regular at the origin. The polynomials grow at infinity, but the weight e^{-x} makes their squared norms finite.

Laguerre equation, polynomials, and orthogonality (cont.)

Set $y = \sum_{k=0}^{\infty} a_k x^k$. Then

$$\begin{aligned}
y' &= \sum_{k=0}^{\infty} (k+1) a_{k+1} x^k, \\
xy'' &= \sum_{k=0}^{\infty} k(k+1) a_{k+1} x^k \\
&\qquad \left(\text{from } y'' = \sum (k+2)(k+1) a_{k+2} x^k, \text{ times } x, \text{ reindex}\right), \\
-xy' &= -\sum_{k=0}^{\infty} k a_k x^k, \\
ny &= \sum_{k=0}^{\infty} n a_k x^k.
\end{aligned}$$

Collecting the coefficient of x^k in $xy'' + (1-x)y' + ny$ gives

$$k(k+1)a_{k+1} + (k+1)a_{k+1} + (n-k)a_k = (k+1)^2 a_{k+1} + (n-k)a_k.$$

Setting every coefficient to zero yields the one-step recurrence

$$a_{k+1} = -\frac{n-k}{(k+1)^2}\, a_k, \qquad k = 0, 1, 2, \ldots. \tag{6.5.2}$$

At $k = n$ one has $a_{n+1} = 0$, so the series terminates as a polynomial of degree n. With the conventional choice $a_0 = 1$, iterating (6.5.2) gives

$$a_k = (-1)^k \frac{n(n-1)\cdots(n-k+1)}{(k!)^2} = (-1)^k \frac{1}{k!}\binom{n}{k},$$

and therefore

$$L_n(x) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{x^k}{k!}. \tag{6.5.3}$$

The first few:

$$L_0 = 1, \quad L_1 = 1 - x, \quad L_2 = 1 - 2x + \tfrac{1}{2}x^2, \quad L_3 = 1 - 3x + \tfrac{3}{2}x^2 - \tfrac{1}{6}x^3.$$

Step 2: Sturm–Liouville form and off-diagonal orthogonality. Multiplying (6.5.1) by e^{-x} produces the SL form $\left(xe^{-x}y'\right)' + ne^{-x}y = 0$. The same boundary-vanishing argument as for Hermite (polynomials and their derivatives are finite at zero, and e^{-x} dominates their growth at infinity) gives, for $n \neq m$,

$$\int_0^\infty L_n(x) L_m(x) e^{-x}\, dx = 0. \tag{6.5.4}$$

Step 3: generating function and the diagonal norm. Define

$$g(x,t) := \frac{1}{1-t} \exp\left(-\frac{xt}{1-t}\right) = \sum_{n=0}^{\infty} L_n(x)\, t^n, \qquad |t| < 1. \tag{6.5.5}$$

The first equality is the definition of g; the second is a claim about the polynomials (6.5.3), and we prove it now. Expand the exponential and absorb the prefactor:

$$\frac{1}{1-t} \exp\left(-\frac{xt}{1-t}\right) = \frac{1}{1-t} \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{-xt}{1-t}\right)^k = \sum_{k=0}^{\infty} \frac{(-x)^k}{k!} \frac{t^k}{(1-t)^{k+1}}.$$

The binomial series from Chapter 1 gives

$$(1-t)^{-(k+1)} = \sum_{j=0}^{\infty} \binom{k+j}{k} t^j, \qquad |t| < 1.$$

Laguerre equation, polynomials, and orthogonality (cont.)

So the power t^n is produced by the pairs with $j = n - k$ and $0 \le k \le n$, each carrying the factor $\binom{k+(n-k)}{k} = \binom{n}{k}$:

$$[t^n]\, g(x,t) = \sum_{k=0}^{n} \frac{(-x)^k}{k!} \binom{n}{k} = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{x^k}{k!} = L_n(x)$$

by (6.5.3). The power series is absolutely convergent for $|t| < 1$, which permits the rearrangement.

Use the double-generating-function method just applied to Hermite. For real s, t near zero, put

$$\alpha = 1 + \frac{s}{1-s} + \frac{t}{1-t} = \frac{1-st}{(1-s)(1-t)} > 0.$$

Then

$$e^{-x} g(x,s) g(x,t) = \frac{e^{-\alpha x}}{(1-s)(1-t)}.$$

Since $\int_0^\infty e^{-\alpha x}\, dx = 1/\alpha$, the integral becomes

$$\int_0^\infty e^{-x} g(x,s) g(x,t)\, dx = \frac{1}{(1-s)(1-t)\alpha} = \frac{1}{1-st}.$$

Laguerre equation, polynomials, and orthogonality (cont.)

Near $s = t = 0$, the exponential decay permits term-by-term integration. Expand both sides. The right side is

$$\frac{1}{1-st} = \sum_{n=0}^{\infty} (st)^n = \sum_{n=0}^{\infty} s^n t^n.$$

The left side is

$$\sum_{m,n=0}^{\infty} \left(\int_0^{\infty} L_m(x) L_n(x) e^{-x}\, dx \right) s^m t^n.$$

Equating coefficients of $s^m t^n$: if $m \neq n$ the integral vanishes (already known); if $m = n$ the coefficient is 1. Therefore the family is orthonormal:

$$\int_0^{\infty} L_m(x) L_n(x) e^{-x}\, dx = \delta_{mn}. \tag{6.5.6}$$

Step 4: three-term recurrence from $\partial_t g$. From the closed form (6.5.5),

$$\frac{\partial g}{\partial t} = g \cdot \frac{1-t-x}{(1-t)^2},$$

or equivalently

$$(1-t)^2\,\partial_t g = (1-t-x)\,g. \tag{6.5.7}$$

Expand both sides in powers of t. With $g = \sum_{n\geq 0} L_n t^n$ one has $\partial_t g = \sum_{n\geq 0}(n+1)L_{n+1}t^n$, so the left side of (6.5.7) is

$$(1-2t+t^2)\sum_{n\geq 0}(n+1)L_{n+1}t^n.$$

The coefficient of t^n on the left is

$$(n+1)L_{n+1} - 2nL_n + (n-1)L_{n-1}$$

(for $n \geq 1$, with the convention $L_{-1} := 0$). On the right, $(1-t-x)\sum L_n t^n$ contributes

$$(1-x)L_n - L_{n-1}.$$

Equating coefficients and rearranging yields

$$(n+1)L_{n+1}(x) = (2n+1-x)L_n(x) - nL_{n-1}(x). \tag{6.5.8}$$

Laguerre equation, polynomials, and orthogonality (cont.)

Check at $n = 1$: $(2)L_2 = (3 - x)L_1 - L_0$ gives $2\left(1 - 2x + \frac{1}{2}x^2\right) = (3 - x)(1 - x) - 1$, which holds identically.

Step 5: Rodrigues formula. The derivative construction is

$$L_n(x) = \frac{e^x}{n!} \frac{d^n}{dx^n}\left(x^n e^{-x}\right). \tag{6.5.9}$$

For $n = 2$, two product-rule steps give

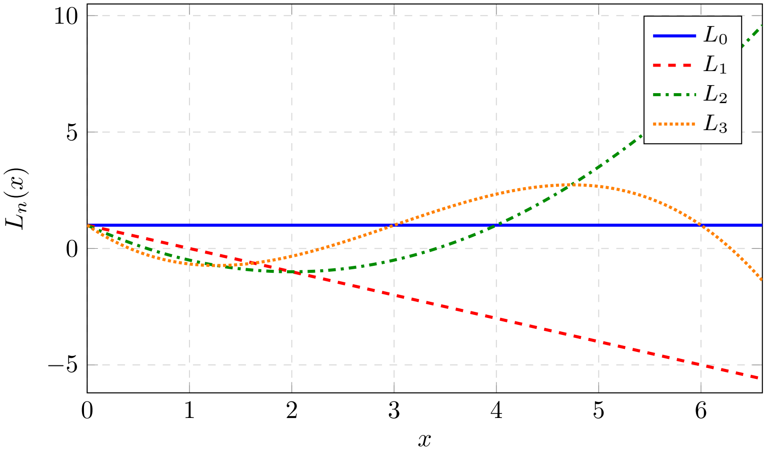
$$\frac{d}{dx}(x^2 e^{-x}) = (2x - x^2)e^{-x},$$
$$\frac{d^2}{dx^2}(x^2 e^{-x}) = (2 - 4x + x^2)e^{-x},$$
$$L_2(x) = \frac{1}{2}(2 - 4x + x^2) = 1 - 2x + \frac{x^2}{2}.$$

For general n, the product rule distributes k derivatives to e^{-x} and $n - k$ to x^n, giving

$$\frac{e^x}{n!} \sum_{k=0}^{n} \binom{n}{k} \frac{n!}{k!} x^k (-1)^k e^{-x} = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{x^k}{k!} = L_n(x).$$

Thus the derivative formula reproduces exactly the series already obtained.

Shapes of the first Laguerre polynomials



All standard Laguerre polynomials satisfy $L_n(0) = 1$. The weight e^{-x} makes them natural on $[0, \infty)$.

Useful formulas for the Laguerre polynomial

differential equation	$xL_n'' + (1-x)L_n' + nL_n = 0$
series / first polynomials	$L_n(x) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{x^k}{k!}$
	$L_0 = 1,\ L_1 = 1-x,\ L_2 = 1-2x+\frac{1}{2}x^2$
generating function	$\frac{1}{1-t}\exp\left(-\frac{xt}{1-t}\right) = \sum_{n=0}^{\infty} L_n(x)t^n$
recurrence	$(n+1)L_{n+1} = (2n+1-x)L_n - nL_{n-1}$
orthonormality	$\int_0^\infty L_m L_n\, e^{-x}\, dx = \delta_{mn}$
Rodrigues	$L_n = \frac{e^x}{n!}\frac{d^n}{dx^n}(x^n e^{-x})$

Generating series requires $|t| < 1$.

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Definition through $\cos n\theta$

The Chebyshev polynomial of the first kind is defined by the elementary identity

$$T_n(\cos\theta) = \cos(n\theta). \tag{6.6.1}$$

That is, set $x = \cos\theta$ and read off a polynomial in x:

$$T_0(x) = 1, \qquad T_1(x) = x, \qquad T_2(x) = 2x^2 - 1, \qquad T_3(x) = 4x^3 - 3x.$$

(For T_2: $\cos 2\theta = 2\cos^2\theta - 1$. For T_3: $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$.)
Recurrence from angle addition. Adding $\cos((n+1)\theta) = \cos(n\theta)\cos\theta - \sin(n\theta)\sin\theta$ and $\cos((n-1)\theta) = \cos(n\theta)\cos\theta + \sin(n\theta)\sin\theta$ cancels the sines:

$$T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x).$$

Thus

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x), \qquad n \geq 1, \tag{6.6.2}$$

with $T_0 = 1$, $T_1 = x$. Each step raises the degree by one: the leading term of $2xT_n$ cannot cancel with T_{n-1}. Starting with leading coefficient 1 for T_1, it doubles at every step, so T_n has leading coefficient 2^{n-1} for $n \geq 1$.

Definition through $\cos n\theta$ (cont.)

The endpoints follow by setting $\theta = 0, \pi$: $T_n(1) = 1$ and $T_n(-1) = (-1)^n$. Replacing θ by $\pi - \theta$ changes x to $-x$, so

$$T_n(-x) = \cos(n\pi - n\theta) = (-1)^n T_n(x).$$

Generating function from the recurrence. A single power series packages all the T_n at once. Put

$$G(x, t) := \sum_{n=0}^{\infty} T_n(x)\, t^n.$$

For $x \in [-1, 1]$, $|T_n(x)| \le 1$, so comparison with $\sum |t|^n$ allows regrouping when $|t| < 1$. Multiply (6.6.2) by t^{n+1} and sum over $n \ge 1$, which is the full range in which the recurrence holds:

$$\sum_{n=1}^{\infty} T_{n+1} t^{n+1} = 2xt \sum_{n=1}^{\infty} T_n t^n - t^2 \sum_{n=1}^{\infty} T_{n-1} t^{n-1}.$$

Each sum is G with a few terms removed or added. Using $T_0 = 1$ and $T_1 = x$,

$$\sum_{n=1}^{\infty} T_{n+1} t^{n+1} = \sum_{k=2}^{\infty} T_k t^k = G - 1 - xt, \qquad \sum_{n=1}^{\infty} T_n t^n = G - 1,$$

$$\sum_{n=1}^{\infty} T_{n-1} t^{n-1} = \sum_{k=0}^{\infty} T_k t^k = G.$$

So $G - 1 - xt = 2xt\,(G-1) - t^2 G$; collecting the G terms on the left and the rest on the right gives $\left(1 - 2xt + t^2\right) G = 1 + xt - 2xt = 1 - xt$, that is

$$\sum_{n=0}^{\infty} T_n(x)\, t^n = \frac{1 - xt}{1 - 2xt + t^2}, \qquad |t| < 1, \quad -1 \le x \le 1. \tag{6.6.3}$$

Differential equation via the chain rule. Set $x = \cos\theta$ and $Y(\theta) := T_n(\cos\theta) = \cos(n\theta)$, so $Y''(\theta) = -n^2\cos(n\theta) = -n^2 Y(\theta)$. Chain rule with $dx/d\theta = -\sin\theta$ gives

$$\frac{d}{d\theta} = -\sin\theta\,\frac{d}{dx}, \qquad \frac{dY}{d\theta} = -\sin\theta\; T_n'(x).$$

Definition through $\cos n\theta$ (cont.)

Differentiate again:

$$\begin{aligned}
\frac{d^2 Y}{d\theta^2} &= \frac{d}{d\theta}\big(-\sin\theta\, T_n'(x)\big) \\
&= -\cos\theta\, T_n'(x) - \sin\theta \cdot \Big(-\sin\theta\, T_n''(x)\Big) \\
&= -\cos\theta\, T_n'(x) + \sin^2\theta\, T_n''(x).
\end{aligned}$$

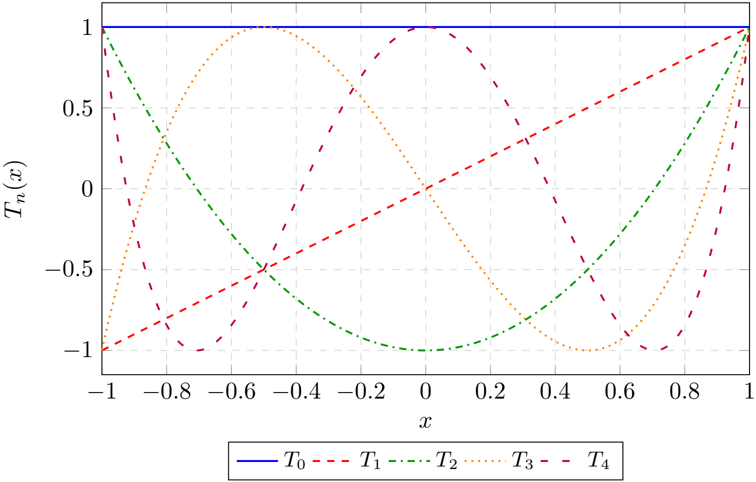
Substitute $\cos\theta = x$ and $\sin^2\theta = 1 - x^2$, and use $Y'' = -n^2 Y = -n^2 T_n$:

$$(1 - x^2)\, T_n''(x) - x T_n'(x) = -n^2 T_n(x).$$

Rearranging yields the Chebyshev equation

$$(1 - x^2)\, T_n''(x) - x T_n'(x) + n^2 T_n(x) = 0. \tag{6.6.4}$$

Shapes of the first Chebyshev polynomials



$T_n(\cos\theta) = \cos(n\theta)$ keeps every T_n between -1 and 1 on $[-1, 1]$.

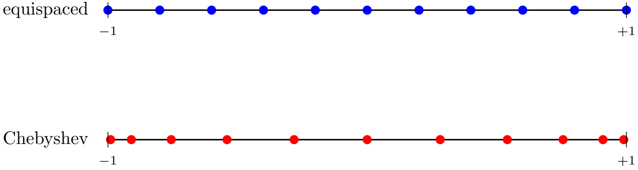
Chebyshev nodes and orthogonality

Zeros (nodes). For $n \geq 1$ and $x = \cos\theta \in [-1, 1]$,

$$T_n(x) = 0 \iff \cos(n\theta) = 0 \iff \theta_k = \frac{(2k-1)\pi}{2n}, \quad k = 1, \ldots, n.$$

$$x_k = \cos\left(\frac{(2k-1)\pi}{2n}\right), \qquad k = 1, \ldots, n. \tag{6.6.5}$$

The angles are equally spaced, but $x = \cos\theta$ crowds the nodes near ± 1. This endpoint clustering is useful for polynomial interpolation.



Chebyshev nodes and orthogonality (cont.)

Orthogonality. Dividing the differential equation by $\sqrt{1-x^2}$ gives

$$\left(\sqrt{1-x^2}\,T_n'\right)' + \frac{n^2}{\sqrt{1-x^2}}\,T_n = 0.$$

Thus the natural weight is $w(x) = 1/\sqrt{1-x^2}$ (up to a constant factor). More intuitively, this weight cancels the change-of-variable factor in $x = \cos\theta$, turning the integral into ordinary cosine orthogonality.
With the weight $(1-x^2)^{-1/2}$, substitute $x = \cos\theta$. Then $dx = -\sin\theta\,d\theta$, and as x runs from -1 to 1, θ runs from π down to 0. Also $\sqrt{1-x^2} = \sqrt{1-\cos^2\theta} = \sin\theta$ for $\theta \in [0,\pi]$. Therefore

$$\begin{aligned}
\int_{-1}^{1} \frac{T_n(x)\,T_m(x)}{\sqrt{1-x^2}}\,dx &= \int_{\pi}^{0} \frac{T_n(\cos\theta)\;T_m(\cos\theta)}{\sin\theta}(-\sin\theta)\,d\theta \\
&= \int_0^{\pi} T_n(\cos\theta)\;T_m(\cos\theta)\,d\theta \\
&= \int_0^{\pi} \cos(n\theta)\cos(m\theta)\,d\theta,
\end{aligned}$$

Chebyshev nodes and orthogonality (cont.)

where the last step is the defining identity $T_k(\cos\theta) = \cos(k\theta)$. For $n \neq m$, use $\cos A \cos B = [\cos(A-B) + \cos(A+B)]/2$:

$$\int_0^\pi \cos(n\theta)\cos(m\theta)\,d\theta = \frac{1}{2}\left[\frac{\sin((n-m)\theta)}{n-m} + \frac{\sin((n+m)\theta)}{n+m}\right]_0^\pi = 0.$$

For $n = m \geq 1$, use $\cos^2(n\theta) = [1 + \cos(2n\theta)]/2$:

$$\int_0^\pi \cos^2(n\theta)\,d\theta = \left[\frac{\theta}{2} + \frac{\sin(2n\theta)}{4n}\right]_0^\pi = \frac{\pi}{2}.$$

For $n = m = 0$, the integrand is 1 and the integral is π. Hence

$$\int_{-1}^1 \frac{T_n(x)\,T_m(x)}{\sqrt{1-x^2}}\,dx = \begin{cases} \pi, & n = m = 0, \\ \pi/2, & n = m \geq 1, \\ 0, & n \neq m. \end{cases} \tag{6.6.6}$$

Useful formulas for the Chebyshev polynomial

definition	$T_n(\cos\theta) = \cos(n\theta)$
differential equation	$(1-x^2)T_n'' - xT_n' + n^2 T_n = 0$
first polynomials	$T_0 = 1,\ T_1 = x,\ T_2 = 2x^2 - 1,\ T_3 = 4x^3 - 3x$
recurrence	$T_{n+1} = 2xT_n - T_{n-1}$
generating function	$\dfrac{1-xt}{1-2xt+t^2} = \sum_{n=0}^{\infty} T_n(x)t^n,\ (6.6.3)$
zeros	$x_k = \cos\left(\dfrac{(2k-1)\pi}{2n}\right)$
orthogonality	$\int_{-1}^{1} \dfrac{T_n T_m}{\sqrt{1-x^2}}\,dx = \begin{cases} \pi, & n = m = 0, \\ \pi/2, & n = m \geq 1, \\ 0, & n \neq m. \end{cases}$

Generating series requires $|t| < 1$.

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Summary: four orthogonal families

Family	Interval	Weight	ODE (sketch)
Legendre P_n	$[-1, 1]$	1	$(1-x^2)y'' - 2xy' + n(n+1)y = 0$
Hermite H_n	$(-\infty, \infty)$	e^{-x^2}	$y'' - 2xy' + 2ny = 0$
Laguerre L_n	$[0, \infty)$	e^{-x}	$xy'' + (1-x)y' + ny = 0$
Chebyshev T_n	$[-1, 1]$	$(1-x^2)^{-1/2}$	$(1-x^2)y'' - xy' + n^2y = 0$

Physical origins (one line each). P_n: axisymmetric Laplace / multipoles; H_n: quantum harmonic oscillator; L_n: its generalized family occurs in the hydrogen radial equation; T_n: $\cos n\theta$ and stable interpolation.

The shared structure is a second-order Sturm–Liouville ODE, a recurrence, and an orthogonality relation on a natural interval and weight. Orthogonality alone only tells us how to *compute* the coefficients of a series; to know that the series actually reproduces f we need the separate property of *completeness*, stated for P_n on the Fourier–Legendre slide. It holds for all four families, as stated on the next slide, and then lets us expand reasonable functions exactly as Fourier series expand in e^{inx}.

Why the expansions converge: completeness

Orthogonality finds the coefficients. Completeness says that the full family can approximate every function with a finite weighted squared norm.
For any of the four families, let ϕ_n denote its polynomials and use the interval I and weight w from the table. Then

$$c_n = \frac{\int_I f(x)\phi_n(x)w(x)\,dx}{\int_I \phi_n(x)^2 w(x)\,dx}, \qquad S_N(x) = \sum_{n=0}^{N} c_n\phi_n(x).$$

The completeness result, used here without proof, is

$$\int_I |f(x) - S_N(x)|^2 w(x)\,dx \longrightarrow 0 \quad \text{whenever} \quad \int_I |f(x)|^2 w(x)\,dx < \infty.$$

This is *mean-square* convergence: the total weighted squared error tends to zero. It does not assert convergence at every individual point.

The common workflow. Choose the family from the differential equation and its interval; identify the weight; compute coefficients by orthogonality; use a finite sum as an approximation.